Memory Notes

Simplified DFI 4.0 Simplified edition - DFI Specification v4.0 (Denali Software)

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List of Waveforms

No waveforms in this document.

1 Simplified Dfi 4 0 Index

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2 Introduction

2.1 What DFI is

The DDR PHY Interface (DFI) is the standard boundary between a memory controller (MC) and a DRAM PHY. The MC schedules commands, manages refresh, and buffers read/write data; the PHY clocks that information out to the DRAM devices, trains the data eyes, and returns read data. DFI splits the problem at a well-defined seam so that MC and PHY can be designed, verified and reused independently.

DFI 4.0 is the DDR4-era revision published in April 2018. It merged the 4.0 Spec Addendum into DFI 3.1 and added support for LPDDR4, DDR4-specific features (bank groups, ACT_n, CA parity, DBI, gear-down), and a set of system-level protocols (PHY master, disconnect, write DQ training, DB training). It remains backward-compatible in spirit: the core control/write/read interface from earlier revisions is still there, wrapped in the new signals.

2.2 Why the seam exists

A memory controller is a traffic manager. It sees banks, rows, columns, refresh counters, and arbitration. A PHY is a timing-and-electrical translator. It sees delay lines, FIFOs, phase alignment, and single-ended vs differential receivers. The two sides operate on different time scales and different abstractions. DFI standardizes the handoff:

- The MC presents commands, address, bank, and chip-select on the control interface.
- The MC presents write data and mask/DBI on the write data interface.
- The PHY returns read data and valid on the read data interface.
- A small set of status and training wires let the PHY report readiness and request calibration.

The MC does not need to know the PHY's internal delay settings; the PHY does not need to know the MC's scheduling policy. Both must agree on DFI timing parameters.

2.3 Edition lineage

DFI revision lineage

DFI revision	Distinctives
DFI 1.0	Initial MC/PHY seam for DDR1/DDR2
DFI 2.0/2.1	DDR3 support, read/write leveling, frequency change, low-power control, 1:2 and 1:4 ratios
DFI 3.0/3.1	DDR4 support (CRC, CA parity, DBI, ACT_n, bank groups), LPDDR3, PHY-requested training, split low-power requests
DFI 4.0	LPDDR4, DB training, per-slice read leveling, read/write data chip selects, write DQ training, PHY master, frequency indicator, disconnect protocol, geardown mode, 3DS/inactive CS support

2.4 What this book covers

This book focuses on the DFI 4.0 signals, timing, and protocols that an andesite-style DDR4/LPDDR4 controller must drive. It is a study condensation, not a replacement for the spec. Every chapter cites the DFI 4.0 section numbers that the material paraphrases. Signal names are used as-is because they are the public vocabulary of the interface.

Source: DFI Specification v4.0 sections 1.0, 2.0, 2.2

3 Organization of the DFI 4.0 Interface

3.1 Sub-interface map

DFI 4.0 groups signals into sub-interfaces. Each group has its own direction, width naming, and timing parameters. The andesite controller drives most of them and ignores a few that are not required at its design point.

DFI 4.0 sub-interface map

Sub-interface	Direction	Purpose	DFI 4.0 section
Control	MC -> PHY	Commands, address, bank, bank-group, CKE, CS, ODT, reset	3.1
Write Data	MC -> PHY	Write data, byte mask/write DBI, write enable, write CS	3.2
Read Data	PHY -> MC (data), MC -> PHY (enables)	Read data, read valid, read DBI, read enable, read CS	3.3
Update	Both	MC-initiated and PHY-initiated update handshakes	3.4
Status	Both	Init, frequency, clock disable, parity, alert	3.5
DFI Training	Both	Gate, read data eye, write leveling, CA, write DQ training	3.6
Low Power Control	MC -> PHY (req), PHY -> MC (ack)	Low-power opportunity requests	3.7

Sub-interface	Direction	Purpose	DFI 4.0 section
Error	PHY -> MC	PHY-detected error indication and info	3.8
DB Training	Both	DDR4 LRDIMM data-buffer training	3.9
PHY Master	PHY -> MC (req), MC -> PHY (ack)	PHY takes control of the DRAM bus	3.10
Geardown	MC -> PHY	DDR4 geardown-mode enable	3.13
LPDDR4 Channels	MC/PHY/system	Channel enable for LPDDR4 combined/independent modes	3.14

3.2 Clocking

DFI is clocked by the DFI clock. In a matched-frequency (1:1) system the DFI clock and the DRAM clock run at the same rate. In frequency-ratio systems the DFI PHY clock is a divided version of the DRAM clock: 1:2 means one DFI clock per two DRAM clocks; 1:4 means one DFI clock per four DRAM clocks.

The andesite design point uses a 1:4 frequency ratio inherited from its predecessor. Four phases of control and write data are transferred per DFI clock; read data is returned as words that map back to those phases.

3.3 Phase, word and cycle replication

DFI 4.0 replicates signals with suffixes that depend on the sub-interface:

- `_pN` (phase): used on control, write data, read enable, parity, and training strobes. Phase 0 may drop the suffix.
- `_wN` (data word): used on read data, read data valid, read data DNV, read data DBI, and DB training response.
- `_aN` (clock cycle): used on `dfi_alert_n`.

For a 1:4 system, `_p0` through `_p3` carry the four DRAM-clock phases of each DFI clock. The MC may issue commands on any phase; the PHY must accept commands on any phase. For read data, `_w0` through `_w3` carry the corresponding data words.

3.4 Slices and channels

A **slice** is an independent data-logic component in the PHY. DFI minimizes slice-to-slice communication and replicates data-control signaling for point-to-point MC-PHY connectivity. In andesite, slices correspond to x8 or x16 data groups; training enables like `dfi_rdlvl_en` are one bit per slice.

LPDDR4 channels are a DFI 4.0 concept. LPDDR4 devices may be configured as two independent 16-bit channels or one combined 32-bit channel. The `phychannel_en` parameter tells the PHY which channels are enabled. The andesite design point uses one DFI bus per channel at x16 width.

Source: DFI Specification v4.0 sections 3.0, 3.11, 3.14, 4.9

4 Acronyms

Acronyms used in this book

Term	Meaning
ACT	Activate command
BG	Bank group (DDR4)
CA	Command/Address bus
CRC	Cyclic redundancy check
CS	Chip select
DB	Data buffer (LRDIMM)
DBI	Data bus inversion
DFI	DDR PHY Interface
DNV	Data not valid
DQ	Data pin group (data + mask/DBI + strobe)
DQS	Data strobe
FGR	Fine granularity refresh
LPDDR	Low-power DDR
MC	Memory controller
MPC	Multi-purpose command (LPDDR4)
MR	Mode register
MRR	Mode register read
MRS	Mode register set
MPR	Multi-purpose register (DDR4)
ODT	On-die termination
PHY	Physical plane (the DRAM interface block)
PRE	Precharge command
RD	Read command
REF	Refresh command
SDR	Single data rate

Term	Meaning
SR	Self-refresh
WR	Write command
3DS	Three-dimensional stacked DRAM

Source: DFI Specification v4.0 section 2.2

5 Control Interface

The control interface carries address, command and control signals from the MC to the PHY. The PHY is responsible for preserving bit ordering and presenting the signals to the DRAM with the delay defined by `tctrl_delay`. In frequency-ratio systems the control bus is phase-replicated with `_pN` suffixes; the MC may issue commands on any phase.

Some signals are DRAM-type-specific. `dfi_reset_n` is used only by DDR3, DDR4 and LPDDR4. `dfi_odt` is used by DDR2, DDR3, DDR4 and LPDDR3. For LPDDR2/3/4 the command encoding pins (`dfi_act_n`, `dfi_ras_n`, `dfi_cas_n`, `dfi_we_n`), bank (`dfi_bank`), bank-group (`dfi_bg`) and chip-ID (`dfi_cid`) must be held constant; the CA bus is mapped onto `dfi_address`.

5.1 Control signals

DFI 4.0 control signals

Signal	Width	Default	Direction	What it does
`dfi_act_n`	1 bit	-	MC -> PHY	Activate select; part of DRAM command encoding with ras/cas/we. Asserted for DDR4 ACT; polarity defines command.
`dfi_address`	DFI Address Width	-	MC -> PHY	Address bus. DDR4: column + lower row bits (A[13:0]); A[16:14] carried on ras/cas/we during ACT. LPDDR2/3: maps 20-bit DDR CA bus. LPDDR4: 6-bit SDR CA.
`dfi_bank`	DFI Bank Width	-	MC -> PHY	Bank address. PHY preserves bit ordering. Constant for LPDDR2/3/4.
`dfi_bg`	DFI Bank Group Width	-	MC -> PHY	DDR4 bank group.

Signal	Width	Default	Direction	What it does
				Constant for LPDDR2/3/4.
dfi_cas_n	DFI Control Width	0x1	MC -> PHY	Column address strobe; command encoding. During DDR4 ACT it carries row address bit A15.
dfi_cid	DFI Chip ID Width	0	MC -> PHY	Chip ID for 3D stacked (3DS) devices. Initializes to 0. Constant for LPDDR2/3/4.
dfi_cke	DFI CKE Width	0x0/0x1	MC -> PHY	Clock enable. MC drives in all phases. For LPDDR3/4 also enables DRAM output drivers during CA training.
dfi_cs	DFI Chip Select Width	0x1	MC -> PHY	Chip select; polarity matches memory signal. During LPDDR3/4 CA training it carries the

Signal	Width	Default	Direction	What it does
				calibration command on the trained rank.
dfi_odt	DFI ODT Width	0x0	MC -> PHY	On-die termination control. MC drives in all phases.
dfi_ras_n	DFI Control Width	0x1	MC -> PHY	Row address strobe; command encoding. During DDR4 ACT it carries row address bit A16.
dfi_reset_n	DFI Reset Width	0x0/0x1	MC -> PHY	DRAM reset. PHY preserves bit ordering.
dfi_we_n	DFI Control Width	0x1	MC -> PHY	Write enable; command encoding. During DDR4 ACT it carries row address bit A14.

5.2 Command encoding on the control pins

For most DRAM types the command is encoded by the classic four pins:

Simplified command encoding on DFI control pins

dfi_act_n	dfi_ras_n	dfi_cas_n	dfi_we_n	Typical command
1	0	1	0	ACT (bank activate)
1	1	0	1	RD (read)
1	1	0	0	WR (write)
1	0	1	1	PRE (precharge)
1	0	0	1	REF (refresh)
0	x	x	x	DDR4 ACT (ACT_n asserted, ras/cas/we carry upper row bits)

DDR4 uses `dfi_act_n` asserted with `dfi_ras_n`, `dfi_cas_n` and `dfi_we_n` carrying row address bits A16, A15 and A14. For LPDDR2/3/4 the command pins are held constant and the CA bus is carried on `dfi_address`.

5.3 LPDDR2/LPDDR3 CA mapping

For LPDDR2 and LPDDR3 the `dfi_address` bus is at least 20 bits wide. The PHY transmits it as a double-data-rate 10-bit CA bus: bits [9:0] on the rising edge and bits [19:10] on the falling edge of the DRAM clock.

5.4 LPDDR4 CA mapping

LPDDR4 uses a 6-bit single-data-rate CA bus. The command is delivered over two consecutive DRAM clock cycles, so a single LPDDR4 command spans two DFI phases. The andesite controller formats this through a dedicated CA submodule in the command formatter.

5.5 Control timing parameters

Control timing parameters

Parameter	Defined by	Description
`tcmd_lat`	MC	DFI PHY clocks from `dfi_cs` assertion until associated CA signals are driven (LPDDR4 CS-to-CA delay).
`tctrl_delay`	PHY	DFI clocks from a control signal change to the change reaching the DRAM interface.

Source: DFI Specification v4.0 sections 3.1, 4.2, Table 5, Table 6

6 Write Data and Read Data Interfaces

The write data interface moves write data and mask/DBI from MC to PHY. The read data interface returns read data, data valid, and read DBI from PHY to MC. Both interfaces are phase- or word-replicated in frequency-ratio systems.

6.1 Write data signals

DFI 4.0 write data signals

Signal	Width	Default	Direction	What it does
`dfi_wrdata`	DFI Data Width	-	MC -> PHY	Write data bus. Data appears `tphy_wrdata` cycles after `dfi_wrdata_en` asserts. Carries MC-generated CRC word when `phycrc_mode = 0`.
`dfi_wrdata_cs`	DFI Physical Rank Width	-	MC -> PHY	Write data-path chip select (polarity of `dfi_cs`). Selects target rank for write data or, during write leveling, the rank being trained. New in 4.0.
`dfi_wrdata_en`	DFI Data Enable Width	0x0	MC -> PHY	Write data/mask enable. Ideally one bit per PHY data slice; all bits identical so

Signal	Width	Default	Direction	What it does
				widths may differ.
dfi_wrdata_mask	DFI Data Width / 8	-	MC -> PHY	Byte mask or write DBI. Same timing as dfi_wrdata. Bit N masks dfi_wrdata[8N+7:8N]. When DBI is enabled and phydbi_mode = 0, this bus carries write DBI instead of mask.

6.1.1 Write DBI and write CRC

Write DBI reuses dfi_wrdata_mask. There is no separate dfi_wrdata_dbi signal in DFI 4.0. If the PHY handles DBI (phydbi_mode = 1) the MC presents plain data and mask; if the MC handles DBI (phydbi_mode = 0) the mask bus is repurposed to carry the DBI pattern.

Write CRC may be generated by the MC (phycrc_mode = 0) or by the PHY (phycrc_mode = 1). When the MC generates CRC, the CRC word is appended on dfi_wrdata after the data burst.

6.2 Read data signals

DFI 4.0 read data signals

Signal	Width	Default	Direction	What it does
`dfi_rddata`	DFI Data Width	-	PHY -> MC	Read data bus. Valid within `tphy_rdlat` cycles after `dfi_rddata_en` asserts. Word-suffixed `_wN`.
`dfi_rddata_cs`	DFI Physical Rank Width	inactive	MC -> PHY	Read data-path chip select (polarity of `dfi_cs`). Selects target rank for read data or, during read training, the rank being trained. New in 4.0.
`dfi_rddata_dbi`	DFI DBI Width	-	PHY -> MC	Read DBI. Sent with `dfi_rddata`; valid only when `phydbi_mode = 0`. Word-suffixed `_wN`.
`dfi_rddata_dnv`	DFI Data Width / 8	0x0	PHY -> MC	Data-not-valid per byte (LPDDR2).

Signal	Width	Default	Direction	What it does
				Same timing as `dfi_rddata_valid`. Word-suffixed `_wN`.
`dfi_rddata_en`	DFI Data Enable Width	0x0	MC -> PHY	Read data enable. Asserted `trddata_en` cycles after read command and held for expected contiguous read data.
`dfi_rddata_valid`	DFI Read Data Valid Width	0x0	PHY -> MC	Per-slice read data valid. One-to-one response to `dfi_rddata_en` within `tphy_rdlat`. Word-suffixed `_wN`.

6.2.1 Read DBI

Read DBI has a dedicated `dfi_rddata_dbi` bus. The PHY returns it alongside `dfi_rddata` when DBI is enabled and `phydbi_mode = 0`. If `phydbi_mode = 1` the PHY inverts data internally and the `dfi_rddata_dbi` signal is not used.

6.3 Programmable parameters for data path

Data-path programmable parameters

Parameter	Defined by	Description
`phycrc_mode`	PHY	`0` = CRC generation/validation in MC; `1` = in PHY.
`phydbi_mode`	PHY	`0` = DBI generation and data inversion in MC; `1` = in PHY.
`dfidata_bit_enable`	MC/PHY	Defines valid data bits on `dfi_wrdata` and `dfi_rddata`. Replaces the removed `dfi_data_byte_disable` signal.

6.4 Key data-path timing parameters

Data-path timing parameters

Parameter	Defined by	Description
`tphy_wrlat`	PHY	DFI PHY clocks from write command to `dfi_wrdata_en` assertion.
`tphy_wrdata`	PHY	DFI PHY clocks from `dfi_wrdata_en` assertion to write data driven on `dfi_wrdata`.
`twrdata_delay`	System	DFI clocks from `dfi_wrdata_en` assertion to write data transfer completion on DRAM bus.
`trddata_en`	System	DFI PHY clocks from read command to `dfi_rddata_en` assertion.
`tphy_rdlat`	PHY	Max DFI PHY clocks from `dfi_rddata_en` assertion to each corresponding `dfi_rddata_valid` bit.
`tphy_rdcslat`	PHY	DFI PHY clocks from read command to assertion of associated `dfi_rddata_cs`.
`tphy_rdcsgap`	PHY	Min additional DFI PHY clocks between commands when changing target rank on `dfi_rddata_cs`.
`tphy_wrcslat`	PHY	DFI PHY clocks from write command to assertion of associated

Parameter	Defined by	Description
		`dfi_wrdata_cs`.
`tphy_wrcsgap`	PHY	Min additional DFI PHY clocks between commands when changing target rank on `dfi_wrdata_cs`.
`tphy_crcmax_lat`	System	Max DFI PHY clocks from `dfi_wrdata_en` (CRC word cycle) to associated CRC error on `dfi_alert_n`.
`tphy_crcmin_lat`	System	Min DFI PHY clocks for the same path.

Source: DFI Specification v4.0 sections 3.2, 3.3, 4.4, 4.5, 4.6, Tables 7-12

7 Update, Status, Error and Low-Power Interfaces

These interfaces handle housekeeping: update handshakes, initialization and frequency signaling, error reporting, and low-power opportunity requests.

7.1 Update interface

The update interface lets either side force a brief idle window on the DFI bus so that the PHY can update internal delay settings or the MC can update its own state.

Update interface signals

Signal	Width	Default	Direction	What it does
`dfi_ctrlupd_req`	1 bit	0x0	MC -> PHY	MC-initiated update request. Asserted min `tctrlupd_min` / max `tctrlupd_max` cycles. Required immediately before a self-refresh exit command in DFI 4.0.
`dfi_ctrlupd_ack`	1 bit	0x0	PHY -> MC	MC-initiated update acknowledge. PHY may ignore; if acknowledging, must assert before `tctrlupd_min` expires. DFI bus stays idle while asserted.
`dfi_phyupd_req`	1 bit	0x0	PHY -> MC	PHY-initiated update request. DFI

Signal	Width	Default	Direction	What it does
				bus must idle while asserted.
dfi_phyupd_ack	1 bit	0x0	MC -> PHY	PHY-initiated update acknowledge. MC must assert within tphyupd_resp and hold while dfi_phyupd_req is asserted.
dfi_phyupd_type	2 bits	-	PHY -> MC	Selects one of four PHY update time classes (tphyupd_type0..3). Constant while request asserted.

7.1.1 Update timing parameters

Update timing parameters

Parameter	Defined by	Description
tctrlupd_interval	MC	Max DFI clocks the MC may wait between dfi_ctrlupd_req assertions.
tctrlupd_min / tctrlupd_max	MC	Min/max cycles dfi_ctrlupd_req can be asserted.
tphyupd_resp	PHY	Max DFI clocks from dfi_phyupd_req

Parameter	Defined by	Description
		assertion to `dfi_phyupd_ack` assertion.
`tphyupd_type0..3`	PHY	Max cycles `dfi_phyupd_req` may stay asserted after ack for each type.

7.2 Status interface

The status interface carries initialization, frequency, clock-disable and parity/alert signals.

Status interface signals

Signal	Width	Default	Direction	What it does
`dfi_alert_n`	DFI Alert Width	0x1	PHY -> MC	CRC or CA parity error indicator. Pulse width mirrors DRAM subsystem error signal. Cycle-suffixed `_aN`.
`dfi_dram_clk_disable`	DFI DRAM Clk Disable Width	0x0	MC -> PHY	When active, PHY must hold DRAM clocks constant. Timing via `tdram_clk_disable` / `tdram_clk_enable`.
`dfi_freq_ratio`	2 bits	-	MC -> PHY	Frequency ratio indicator: `00`=1:1, `01`=1:2, `10`=1:4, `11`=reserved. Defined at init, then constant.
`dfi_frequency`	5 bits	-	MC -> PHY	Operating-

Signal	Width	Default	Direction	What it does
				frequency indicator. Up to 32 system-defined encodings. New in 4.0.
`dfi_init_complete`	1 bit	0x0	PHY -> MC	PHY ready to accept DFI transactions. All command/status signals hold defaults until it asserts. De-assertion acknowledges a frequency change.
`dfi_init_start`	1 bit	0x0/0x1	MC -> PHY	At init: indicates `dfi_freq_ratio` and `dfi_frequency` are valid. During operation: requests a frequency change.
`dfi_parity_in`	1 bit	0x0/0x1	MC -> PHY	Command parity, one-to-one with each DFI command. Covers

Signal	Width	Default	Direction	What it does
				address, bank, bank-group, act_n, cas_n, cid, ras_n, we_n. Phase-suffixed _pN.

7.2.1 Status timing parameters

Status timing parameters

Parameter	Defined by	Description
tdram_clk_disable	PHY	DFI clocks from dfi_dram_clk_disable assertion until DRAM clock holds low at PHY-DRAM boundary.
tdram_clk_enable	PHY	DFI clocks from dfi_dram_clk_disable de-assertion to first valid DRAM clock rising edge.
tinit_complete	PHY	Max DFI clocks from dfi_init_start de-assertion to dfi_init_complete re-assertion during frequency change.
tinit_complete_min	PHY	Min DFI clocks before dfi_init_complete can be driven after a previous command or training event.
tinit_start	MC	DFI clocks from dfi_init_start assertion until PHY must de-assert dfi_init_complete to

Parameter	Defined by	Description
		accept frequency change.
`tinit_start_min`	PHY	Min DFI clocks before `dfi_init_start` can be driven after a previous command or training event.
`tparin_lat`	MC	DFI PHY clocks between DFI command assertion and associated `dfi_parity_in`.
`tphy_paritylat`	PHY	Max DFI clocks between `dfi_parity_in` and associated `dfi_alert_n` return (min 0x4).

7.2.2 Programmable status parameters

Status programmable parameters

Parameter	Defined by	Description
`dfidata_bit_enable`	MC/PHY	Valid data bits on `dfi_wrdata` and `dfi_rddata`. Replaces removed `dfi_data_byte_disable`.
`phyfreq_range`	PHY	Number of `dfi_frequency` values the PHY supports (1-32).

7.3 Error interface

Error interface signals

Signal	Width	Default	Direction	What it does
`dfi_error`	DFI Error Width	0x0	PHY -> MC	PHY-detected error condition (DRAM errors, DLL lock loss, read DQS error, etc.). Not phased.
`dfi_error_info`	DFI Error Width x 4	0x0	PHY -> MC	Additional error-source information. Valid only when `dfi_error` asserted. Partially spec-defined, extensible with design-specific codes.

Error timing parameter

Parameter	Defined by	Description
`terror_resp`	PHY	Max DFI clocks from affected DFI bus transaction(s) to `dfi_error` assertion.

7.4 Low-power control interface

Low-power control interface signals

Signal	Width	Default	Direction	What it does
`dfi_lp_ack`	1 bit	0x0	PHY -> MC	Low-power acknowledge. PHY may ignore; if acknowledging, asserts within `tlp_resp` and de-asserts within `tlp_wakeup` after request de-asserts.
`dfi_lp_ctrl_req`	1 bit	0x0	MC -> PHY	Low-power request for the control interface: no more commands will be sent.
`dfi_lp_data_req`	1 bit	0x0	MC -> PHY	Low-power request for the data interface: no more data will be sent.
`dfi_lp_wakeup`	4 bits	-	MC -> PHY	Requested wakeup time, 16 encodings from 16 cycles (`0000`) doubling up

Signal	Width	Default	Direction	What it does
				to 262144 (1110) and unlimited (1111). May increase but never decrease after acknowledge.

Low-power timing parameters

Parameter	Defined by	Description
tlp_resp	MC	Max DFI clocks from low-power request assertion to dfi_lp_ack assertion (min 1, max 7; recommended 7).
tlp_wakeup	MC	Target max DFI clocks that dfi_lp_ack may stay asserted after request de-asserts (min 16).

Source: DFI Specification v4.0 sections 3.4, 3.5, 3.7, 3.8, Tables 13-17, 21, 22

8 Training, DB Training, PHY Master, Geardown and LPDDR4 Channels

DFI 4.0 adds several specialized interfaces beyond the core control/data path. This chapter covers training (gate, read data eye, write leveling, CA, write DQ), DB training for LRDIMM, the PHY master protocol, geardown mode, and LPDDR4 channel configuration.

8.1 DFI training interface

DFI 4.0 supports four training operations: gate training, read data eye training, write leveling, and CA training. Write DQ training is new in 4.0. Training can run in DFI Training Mode (MC/PHY handshake over DFI) or PHY Independent Mode (PHY runs training on its own). The andesite design point uses DFI Training Mode.

8.1.1 Training signals

DFI training interface signals

Signal	Width	Default	Direction	What it does
`dfi_calvl_ca_sel`	Defined by `tcalvl_ca_sel`	0x0	MC -> PHY	CA foreground pattern strobe; indicates when foreground pattern should be transmitted PHY-to-DRAM.
`dfi_calvl_capture`	DFI CA Training MC I/F Width	0x0	MC -> PHY	One-DFI-clock pulse initiating capture of CA bus value from DQ bus.
`dfi_calvl_data`	7 bits per slice	-	MC -> PHY	CA VREF training value to transfer to DRAM, per slice.
`dfi_calvl_done`	1 bit	0x0	MC -> PHY	MC has finished iterating VREF values; PHY responds with

Signal	Width	Default	Direction	What it does
				dfi_calvl_resp.
dfi_calvl_en	Per slice	0x0	MC -> PHY	CA training logic enable per slice; assertion initiates CA training and acknowledges dfi_calvl_req.
dfi_calvl_req	Per slice	0x0	PHY -> MC	PHY-initiated CA training request per slice.
dfi_calvl_resp	2 bits	0x0	PHY -> MC	CA training response: 00 in-process, 01 LPDDR4 VREF sequence done / LPDDR3 reserved, 10 LPDDR3 change-segment / LPDDR4 reserved, 11 complete.
dfi_calvl_result	1 bit	0x0	PHY -> MC	VREF comparison result: 1 = current value better than previous best.

Signal	Width	Default	Direction	What it does
dfi_calvl_strobe	Per slice	0x0	MC -> PHY	CA VREF training strobe per slice; each DFI clock of assertion generates one DQS pulse.
dfi_lvl_pattern	4 bits x read-training width	0x0	MC -> PHY	Read-training pattern select. For DDR4/LPDDR2/3 defines pattern directly; for LPDDR4 indexes a programmed table.
dfi_lvl_periodic	Leveling width	0x0	MC -> PHY	Global training length: 0 = full/long, 1 = periodic/short.
dfi_phy_calvl_cs	DFI Physical Rank Width	inactive	PHY -> MC	CA training target chip select.
dfi_phy_rdlvl_cs	Rank x read-training width	inactive	PHY -> MC	Read data eye training target chip select.
dfi_phy_rdlvl_gate_cs	Rank x read-training width	inactive	PHY -> MC	Gate training target chip select.

Signal	Width	Default	Direction	What it does
dfi_phy_wdqlvl_cs	Rank x write-DQ width	inactive	PHY -> MC	Write DQ training target chip select.
dfi_phy_wrlvl_cs	Rank x write-leveling width	inactive	PHY -> MC	Write leveling target chip select.
dfi_rdlvl_done	Per slice	0x0	MC -> PHY	Read data eye training: MC finished iterating training values.
dfi_rdlvl_en	Per slice	0x0	MC -> PHY	Read data eye training enable per slice; acknowledges dfi_rdlvl_req.
dfi_rdlvl_gate_en	Per slice	0x0	MC -> PHY	Gate training enable per slice; acknowledges dfi_rdlvl_gate_req.
dfi_rdlvl_gate_req	Per slice	0x0	PHY -> MC	PHY-initiated gate training request per slice.
dfi_rdlvl_req	Per slice	0x0	PHY -> MC	PHY-initiated read data eye training request per slice.

Signal	Width	Default	Direction	What it does
dfi_rdlvl_resp	Read response width	0x0	PHY -> MC	Read training status: 00 in-process, 01 sequence done (data-eye only), 10 reserved, 11 complete.
dfi_wdqlvl_done	Per slice	0x0	MC -> PHY	Write DQ training: MC finished iterating VREF values. New in 4.0.
dfi_wdqlvl_en	Per slice	0x0	MC -> PHY	Write DQ training enable per slice. New in 4.0.
dfi_wdqlvl_req	Write-DQ width	0x0	PHY -> MC	PHY write DQ training request. New in 4.0.
dfi_wdqlvl_resp	Write-DQ response width	0x0	PHY -> MC	Write DQ training response. New in 4.0.
dfi_wdqlvl_result	1 bit	0x0	PHY -> MC	Write DQ VREF comparison result. New in 4.0.
dfi_wrlvl_en	Per slice	0x0	MC -> PHY	Write leveling enable per slice;

Signal	Width	Default	Direction	What it does
				acknowledges dfi_wrlvl_req.
dfi_wrlvl_req	Write-leveling width	0x0	PHY -> MC	PHY write leveling request.
dfi_wrlvl_resp	Per slice	0x0	PHY -> MC	Write leveling response: 0 in-process/not active, 1 complete.
dfi_wrlvl_strobe	Per slice	0x0	MC -> PHY	Write leveling strobe per slice; assert count per request set by syswrlvl_strobe_num.

8.1.2 Training timing parameters

Training timing parameters

Parameter	Description
tcalvl_ca_sel	Width generated by combining per-phase PHY inputs for dfi_calvl_ca_sel.
tcalvl_capture	DFI clocks after calibration command on dfi_cs before dfi_calvl_capture pulse.
tcalvl_cc	Min DFI clocks from one calibration command to the next.
tcalvl_cs_ca	PHY DFI clocks from dfi_calvl_ca_sel assertion to dfi_cs.
tcalvl_data	Min DFI PHY clocks dfi_calvl_data must be held for DRAM hold time.

Parameter	Description
tcalvl_en	Min DFI clocks from dfi_calvl_en assertion to dfi_cke de-assertion.
tcalvl_max	Max DFI clocks MC waits for dfi_calvl_resp after asserting dfi_calvl_en.
tcalvl_resp	Max DFI clocks from dfi_calvl_req to MC asserting dfi_calvl_en.
tcalvl_strobe	Min DFI PHY clocks from dfi_calvl_data to dfi_calvl_strobe for DRAM setup.
trdlvl_en	Min DFI clocks from dfi_rdlvl_en/dfi_rdlvl_gate_en to first read/MRR command.
trdlvl_max	Max DFI clocks MC waits for dfi_rdlvl_resp after setting read training enable.
trdlvl_resp	Max DFI clocks from read training request to MC asserting enable.
trdlvl_rr	Min DFI clocks between consecutive read-training read commands.
twdqlvl_en	Min DFI clocks from dfi_wdqlvl_en to first write command.
twdqlvl_max	Max DFI clocks MC waits for dfi_wdqlvl_resp after asserting dfi_wdqlvl_en.
twdqlvl_resp	Max DFI clocks from dfi_wdqlvl_req to MC asserting dfi_wdqlvl_en.
twdqlvl_rw	Min DFI clocks from last read to first write of next calibration commands.
twdqlvl_ww	Min DFI clocks between write commands during write DQ training.
twrlvl_en	Min DFI clocks from dfi_wrlvl_en to first dfi_wrlvl_strobe.

Parameter	Description
twrlvl_max	Max DFI clocks MC waits for dfi_wrlvl_resp after setting dfi_wrlvl_en.
twrlvl_resp	Max DFI clocks from dfi_wrlvl_req to MC asserting dfi_wrlvl_en.
twrlvl_ww	Min DFI clocks from one dfi_wrlvl_strobe to the next.

8.1.3 Training programmable parameters

Training programmable parameters

Parameter	Description
mcrdlvl_slice_group / phyrdlvl_slice_group	Per-slice grouping of dfi_rdlvl_en bits. All-1 reproduces DFI 3.1 behavior; all-0 enables fully independent per-slice leveling.
phycalvl_en	1 = DFI CA training enabled in PHY.
phyrdlvl_en / phyrdlvl_gate_en	PHY supports read data eye / gate training on DFI bus.
phywrlvl_en	PHY supports write leveling on DFI bus.
phywdqlvl_bst	Number of consecutive write bursts per write DQ training sequence.
syswrlvl_strobe_num	Minimum clocks dfi_wrlvl_strobe asserts per write leveling request.

8.2 DB training interface

The DB training interface supports DDR4 LRDIMM data-buffer-to-DRAM training in PHY evaluation mode.

DB training interface signals

Signal	Width	Default	Direction	What it does
`dfi_db_train_en`	Per slice	0x0	MC -> PHY	Enables DB training mode: PHY transfers commands from MC and captures DQ inputs for return on response bus.
`dfi_db_train_resp`	Per slice x DQ per slice	0x0	PHY -> MC	DQ data returned to MC for evaluation during DB training. Word-suffixed `_wN`.

DB training timing parameters

Parameter	Description
`tphy_db_train_en`	PHY DFI clocks after `dfi_db_train_en` asserts until first training command can be issued.
`tphy_db_train_resp`	Max DFI clocks from PHY DQ input to `dfi_db_train_resp` output.

8.3 PHY master interface

The PHY master interface lets the PHY take temporary control of the DRAM bus. This is used for PHY-initiated training or other maintenance that requires the PHY to drive commands directly.

PHY master interface signals

Signal	Width	Default	Direction	What it does
`dfi_phymstr_ack`	1 bit	0x0	MC -> PHY	PHY master acknowledge: MC has placed DRAM in a known state and PHY is master of the DRAM bus.
`dfi_phymstr_cs_state`	DFI Chip Select Width	0x0	PHY -> MC	Per-rank DRAM state when PHY becomes master: `0` = IDLE or self-refresh per `dfi_phymstr_state_sel`; `1` = MC may choose any supported state.
`dfi_phymstr_req`	1 bit	0x0	PHY -> MC	PHY requests control of DFI/DRAM buses.
`dfi_phymstr_state_sel`	1 bit	0x0	PHY -> MC	For CS with `dfi_phymstr_cs_state=0`: `0` = MC must

Signal	Width	Default	Direction	What it does
				place memory in IDLE, 1 = self-refresh.
`dfi_phymstr_type`	2 bits	0x0	PHY -> MC	Selects one of four PHY master time classes (`tphymstr_type0..3`).

PHY master timing parameters

Parameter	Description
`tphymstr_resp`	Max DFI clocks from `dfi_phymstr_req` to `dfi_phymstr_ack`.
`tphymstr_rfsh`	Max DFI clocks PHY needs to forward a refresh command from MC to DRAM while master.
`tphymstr_type0..3`	Max cycles `dfi_phymstr_req` may stay asserted after ack for each type.

8.3.1 Inactive chip selects

DFI 4.0 introduces `syscs_state`, a per-chip-select state: `0` = inactive (unpopulated, uninitialized, powered off, DPD, MPSM), `1` = active. The PHY must not assert training chip-select bits for inactive ranks. This lets a controller and PHY handle partially populated systems without treating empty slots as errors.

8.4 Geardown mode

Geardown mode is a DDR4 feature where the CA bus runs at half the memory clock rate. The MC changes CA signals only every other DFI PHY clock, synchronized by the PHY.

Geardown mode signal

Signal	Width	Default	Direction	What it does
`dfi_geardown_en`	1 bit	0x0	MC -> PHY	When asserted, MC and PHY operate in DDR4 geardown mode. New in 4.0.

Geardown timing parameter

Parameter	Description
`tgeardown_delay`	Delay from `dfi_geardown_en` assertion until PHY is ready to receive commands.

8.5 LPDDR4 channels

LPDDR4 supports two independent 16-bit channels or one combined 32-bit channel.

LPDDR4 channel programmable parameter

Parameter	Description
`phychannel_en`	Which DFI channels are enabled in LPDDR4 combined mode: `01` = ch0, `10` = ch1, `11` = both. Independent mode requires both.

Source: DFI Specification v4.0 sections 3.6, 3.9, 3.10, 3.11, 3.13, 3.14, Tables 18-20, 23-26

9 Reset and Initialization

DFI initialization is a handshake between MC and PHY mediated by `dfi_init_start` and `dfi_init_complete`. The same two signals are reused for frequency change during normal operation.

9.1 Power-up and the init handshake

At power-up the PHY may optionally wait for `dfi_init_start` before asserting `dfi_init_complete`. The MC drives `dfi_freq_ratio` and `dfi_frequency` to their initial values and then asserts `dfi_init_start`. This tells the PHY that the frequency ratio and frequency indicator are valid.

Initialization completes when both `dfi_init_start` and `dfi_init_complete` are asserted simultaneously for at least one DFI clock. After that point the PHY accepts normal DFI transactions and all command/status signals may leave their defaults.

Initialization handshake phases

Phase	MC behavior	PHY behavior
Power-up	Drive `dfi_freq_ratio` and `dfi_frequency` to initial values.	May wait for `dfi_init_start`.
Init request	Assert `dfi_init_start`.	Accept ratio/frequency; prepare DRAM interface.
Init complete	May hold or de-assert `dfi_init_start` after both signals are high.	Assert `dfi_init_complete`.
Normal operation	Issue commands, data, refresh.	Respond to DFI transactions.

9.2 Frequency change

During normal operation the MC requests a frequency change by asserting `dfi_init_start` again. The PHY accepts by de-asserting `dfi_init_complete` within `tinit_start` cycles. If the PHY does not de-assert `dfi_init_complete` within `tinit_start`, the MC must abort the request and release `dfi_init_start`.

While `dfi_init_start` is asserted for a frequency change, `dfi_frequency` must be set to the new legal value and remain unchanged. Once `dfi_init_complete` de-asserts, the MC holds `dfi_init_start` until the frequency change is complete. The PHY re-asserts `dfi_init_complete` within `tinit_complete` cycles after `dfi_init_start` de-asserts.

9.3 Clock disabling during init

When `dfi_init_start` is asserted during initialization, `dfi_dram_clk_disable` must reflect the clocks that are being used. In normal operation this signal can change dynamically to gate the DRAM clocks for power savings.

9.4 Init timing parameters

Initialization timing parameters

Parameter	Description
`tinit_start`	Max DFI clocks from `dfi_init_start` assertion until PHY must de-assert `dfi_init_complete` to accept frequency change.
`tinit_start_min`	Min DFI clocks before `dfi_init_start` can be driven after a previous command or training event.
`tinit_complete`	Max DFI clocks from `dfi_init_start` de-assertion to `dfi_init_complete` re-assertion during frequency change.
`tinit_complete_min`	Min DFI clocks before `dfi_init_complete` can be driven after a previous command or training event.

9.5 Update before self-refresh exit

DFI 4.0 requires a `dfi_ctrlupd_req`/`dfi_ctrlupd_ack` handshake immediately before a self-refresh exit command. This update window guarantees that the PHY has a clean point to apply any pending delay updates before the command stream resumes at full speed.

Source: DFI Specification v4.0 sections 3.5.1, 3.5.5, 4.1, 4.10

10 Training Operations

DFI 4.0 training moves the calibration of read gates, data eyes, write DQS/DQ timing, and CA alignment from the MC into a cooperative MC/PHY protocol. The PHY evaluates DRAM responses and adjusts delay settings; the MC sets up the DRAM, issues training commands, and responds to PHY requests.

10.1 Training modes

Two modes exist:

- **DFI Training Mode:** MC and PHY cooperate through the DFI training handshake. This is the mode described in this book.
- **PHY Independent Mode:** PHY runs training autonomously after initialization. The MC does not participate beyond enabling the capability.

The andesite controller implements DFI Training Mode for the training operations it supports.

10.2 Gate training

Gate training finds the delay at which the read DQS preamble can be reliably captured. The MC asserts `dfi_rdlvl_gate_en` for the target slice(s), issues the read or MRR commands required by the memory type, and the PHY adjusts the gate delay until the DQS window is centered.

When complete, the PHY drives `dfi_rdlvl_resp = 'b11`. The MC then de-asserts `dfi_rdlvl_gate_en`. The `dfi_lvl_pattern` signal selects the pattern for read data eye training but is not used for gate training.

10.3 Read data eye training

Read data eye training centers the read capture clock in the valid data eye. The MC asserts `dfi_rdlvl_en` for the target slice(s) and issues read/MRR commands. The PHY evaluates the data and adjusts read delay.

If the PHY returns `dfi_rdlvl_resp = 'b01`, the current sequence is complete but more sequences remain. The MC de-asserts `dfi_rdlvl_en`, sets up the next sequence (e.g., next `dfi_lvl_pattern` value), and re-asserts `dfi_rdlvl_en`. When the PHY returns `'b11`, training is complete.

For LPDDR4, `dfi_lvl_pattern` is an index into a programmed pattern table rather than a direct pattern encoding.

10.4 Write leveling

Write leveling aligns the write DQS rising edge with the DRAM clock rising edge. The MC asserts `dfi_wrlvl_en`, then asserts `dfi_wrlvl_strobe` for the number of clocks defined by `syswrlvl_strobe_num`. The PHY samples the DQ bus response and adjusts write DQS delay.

When the PHY returns `dfi_wrlvl_resp = 'b1`, write leveling is complete for that slice.

10.5 CA training

CA training is required for LPDDR3 and LPDDR4. The MC asserts `dfi_calvl_en`, drives the calibration command on `dfi_cs`, and uses `dfi_calvl_capture` to tell the PHY when to sample the CA value returned on the DQ bus. For LPDDR4 CA VREF training, the MC drives `dfi_calvl_data`, asserts `dfi_calvl_strobe` to generate DQS pulses, and uses `dfi_calvl_done` and `dfi_calvl_result` to iterate to the best VREF setting.

10.6 Write DQ training

Write DQ training is new in DFI 4.0 and centers the clock in the write DQ eye. The MC asserts `dfi_wdqlvl_en`, issues a sequence of write bursts followed by matching read bursts, and iterates VREF values using `dfi_wdqlvl_done` and `dfi_wdqlvl_result`. The PHY returns `dfi_wdqlvl_resp` status. This training is unused unless both MC and PHY are DFI 4.0 devices.

10.7 DB training

DB training supports DDR4 LRDIMM data-buffer training in PHY evaluation mode. The MC asserts `dfi_db_train_en`, transfers commands through the PHY, and evaluates `dfi_db_train_resp`. This is specialized to LRDIMM systems and is not part of the andesite design point.

10.8 PHY master during training

The PHY master interface can be used when the PHY needs to take control of the DRAM bus for extended calibration. The PHY asserts `dfi_phymstr_req`; the MC places the DRAM in IDLE or self- refresh and asserts `dfi_phymstr_ack`. The MC may still forward refresh commands, which the PHY relays to the DRAM. This is a DFI 4.0 feature and is not required for basic DDR4/LPDDR4 operation.

10.9 Training chip-select renaming

A subtle but important 3.1 -> 4.0 change is the removal of the `_n` suffix on training chip-select signals. DFI 4.0 uses `dfi_phy_wrlvl_cs`, `dfi_phy_rdlvl_cs`, `dfi_phy_rdlvl_gate_cs` and `dfi_phy_calvl_cs`. The polarity is the same as `dfi_cs`; the name simply no longer embeds the active-low assumption.

Source: DFI Specification v4.0 sections 3.6, 3.9, 3.10, 3.12, 4.12, 4.15, 4.16

11 Command Mapping

DFI does not define DRAM commands directly. It defines the wires that carry them. The MC must map each DRAM command onto the correct pattern of `dfi_act_n`, `dfi_ras_n`, `dfi_cas_n`, `dfi_we_n`, `dfi_address`, `dfi_bank`, `dfi_bg`, `dfi_cid`, `dfi_cs` and `dfi_cke`.

11.1 DDR4 command mapping

For DDR4, `dfi_act_n` distinguishes ACT from other commands. During ACT, `dfi_ras_n`, `dfi_cas_n` and `dfi_we_n` carry row address bits A16, A15 and A14. During non-ACT commands they carry the classic RAS/CAS/WE encoding.

Simplified DDR4 command mapping on DFI control wires

DRAM command	dfi_act_n	dfi_ras_n	dfi_cas_n	dfi_we_n	dfi_address	dfi_bank	dfi_bg
ACT	0	A16	A15	A14	Row[13:0]	Bank	Bank group
RD	1	1	0	1	Column + A10=auto-precharge	Bank	Bank group
WR	1	1	0	0	Column + A10=auto-precharge	Bank	Bank group
PRE	1	0	1	1	A10=precharge-all	Bank	Bank group
REF	1	0	0	1	-	-	-
MRS	1	0	0	0	MR data	MR address	-
DES	1	1	1	1	-	-	-

11.2 LPDDR4 CA command mapping

LPDDR4 uses a 6-bit single-data-rate CA bus delivered over two consecutive DRAM clock cycles. The command formatter must hold the command pins constant and drive the CA encoding onto `dfi_address[5:0]` across two phases. The exact two-cycle encoding is defined by JESD209-4; the andesite controller generates it through a dedicated CA submodule.

Because the CA bus is SDR and commands span two cycles, the MC cannot change CA arbitrarily on every phase. The CA valid qualifier marks both cycles of a command.

11.3 LPDDR2/LPDDR3 CA mapping

LPDDR2 and LPDDR3 use a 10-bit double-data-rate CA bus. The MC drives a 20-bit value on `dfi_address`; the PHY transmits bits [9:0] on the rising edge and bits [19:10] on the falling edge. Command pins (`dfi_act_n`, `dfi_ras_n`, `dfi_cas_n`, `dfi_we_n`, `dfi_bank`, `dfi_bg`, `dfi_cid`) are held constant.

11.4 Chip select and rank targeting

`dfi_cs` selects the rank for control commands. DFI 4.0 adds `dfi_wrdata_cs` and `dfi_rddata_cs` to separately identify the target rank on the data path. In a single-rank andesite configuration these are driven constant. In multi-rank or 3DS configurations they follow the rank selected by the command.

11.5 CA parity

DDR4 supports command/address parity. The MC generates parity on `dfi_parity_in` covering `dfi_address`, `dfi_bank`, `dfi_bg`, `dfi_act_n`, `dfi_cas_n`, `dfi_cid`, `dfi_ras_n` and `dfi_we_n`. The PHY returns any detected parity error on `dfi_alert_n`. The `tparin_lat` and `tphy_paritylat` parameters define the timing relationship.

Source: DFI Specification v4.0 sections 3.1, 4.2, 4.3, 4.11, 4.19

12 Write Carriage

The write data interface carries write data, mask or write DBI, and enable signals from MC to PHY. The PHY forwards them to the DRAM after the programmed write latency.

12.1 Write timing overview

For a matched-frequency system, the sequence is:

1. MC issues write command on control interface.
2. After `tphy_wrlat` DFI clocks, MC asserts `dfi_wrdata_en`.
3. After `tphy_wrdata` DFI clocks from `dfi_wrdata_en`, first write data word appears on `dfi_wrdata`.
4. `dfi_wrdata_en` stays asserted for the duration of the burst. `dfi_wrdata_mask` (or write DBI when enabled) travels with the data.

In a 1:4 frequency-ratio system, the control command is issued on one phase and the phase-suffixed write data (`dfi_wrdata_p0`..`p3`) carries four DRAM-clock words per DFI clock.

12.2 Write data enable

`dfi_wrdata_en` is ideally one bit per PHY data slice. All bits are identical in normal operation, so the MC-to-PHY width may differ. The enable must be asserted early enough to satisfy `tphy_wrlat` and must remain asserted for the entire burst.

12.3 Write mask and write DBI

`dfi_wrdata_mask` is one bit per byte of write data. When DBI is disabled, a `1` masks the associated byte. When DBI is enabled and `phydbi_mode = 0`, the same bus carries the DBI pattern: a `1` tells the DRAM to invert the corresponding byte before storing it. There is no separate `dfi_wrdata_dbi` signal in DFI 4.0.

When `phydbi_mode = 1`, the PHY computes and inserts DBI internally; the MC presents plain data and mask.

12.4 Write CRC

Write CRC may be generated in the MC (`phycrc_mode = 0`) or in the PHY (`phycrc_mode = 1`). When the MC generates CRC, the CRC word is appended to the write burst on `dfi_wrdata`. The PHY reports CRC errors on `dfi_alert_n` with timing bounded by `tphy_crcmin_lat` and `tphy_crcmax_lat`.

12.5 Write data chip select

`dfi_wrdata_cs` identifies the target rank for the write data. During write leveling it identifies the rank being trained. The PHY may need additional rank-switching gap (`tphy_wrcsgap`) when the target rank changes between writes.

Source: DFI Specification v4.0 sections 3.2, 4.4, 4.5, 4.5.2, 4.5.3

13 Read Carriage

The read data interface returns read data from PHY to MC. The MC must tell the PHY when to expect data by asserting `dfi_rddata_en`; the PHY returns data and per-slice valid signals after the read latency.

13.1 Read timing overview

For a matched-frequency system:

1. MC issues read command on control interface.
2. After `trddata_en` DFI clocks, MC asserts `dfi_rddata_en` for the expected burst duration.
3. Within `tphy_rdlat` DFI clocks from `dfi_rddata_en`, the PHY asserts `dfi_rddata_valid` and drives `dfi_rddata`.

In a 1:4 frequency-ratio system, `dfi_rddata_en` is phase-suffixed and the returned read data appears as word-suffixed `_w0`..`_w3`.

13.2 Read data valid

`dfi_rddata_valid` is one bit per PHY data slice. It is the one-to-one response to `dfi_rddata_en` within `tphy_rdlat`. The MC uses it to qualify `dfi_rddata`. Stray valid signals without a matching `dfi_rddata_en` are dropped.

13.3 Read DBI

When DBI is enabled and `phydbi_mode = 0`, the PHY returns `dfi_rddata_dbi` alongside `dfi_rddata`. A `1` indicates that the corresponding byte was inverted by the DRAM and must be re-inverted by the receiver. When `phydbi_mode = 1` the PHY handles inversion internally and `dfi_rddata_dbi` is not used.

13.4 Read data chip select

`dfi_rddata_cs` identifies the rank for which read data is expected. During read training it identifies the rank being trained. The PHY may need additional rank-switching gap (`tphy_rdcsgap`) when the target rank changes between reads.

13.5 Data not valid (LPDDR2)

`dfi_rddata_dnv` is an LPDDR2-specific per-byte signal that marks bytes which do not contain valid data. It has the same timing as `dfi_rddata_valid`.

Source: DFI Specification v4.0 sections 3.3, 4.4, 4.6, 4.6.2

14 Wire-Level Walkthroughs

These walkthroughs use a deliberately tiny teaching model so the DFI signals are visible without being lost in real memory geometry.

14.1 The simplified model

- One rank, one x8 data slice.
- 1:4 frequency ratio: four phases (`_p0`..`_p3`) per DFI clock, four read data words (`_w0`..`_w3`) per DFI clock.
- Two banks: B0, B1.
- One row per bank: R0.
- Burst length 8 (BL8), so a single column read returns 8 contiguous data words.
- Timing assumptions (illustrative only):
  - `tphy_wrlat` = 2 DFI clocks
  - `tphy_wrdata` = 1 DFI clock
  - `trddata_en` = 2 DFI clocks
  - `tphy_rdlat` = 3 DFI clocks

The model is unrealistic in size but correct in wire-level behavior.

14.2 Walkthrough 1: activate -> read -> data return

The MC wants to read from bank B0, row R0, column C0.

At DFI clock T0, phase p0, the MC drives the ACT command:

ACT command driven at T0/p0

Signal	Value	Comment
`dfi_cs_p0`	0 (active)	Chip select asserted
`dfi_act_n_p0`	0	ACT form
`dfi_ras_n_p0`	A16 value	Row bit, plus command encoding
`dfi_cas_n_p0`	A15 value	Row bit, plus command encoding
`dfi_we_n_p0`	A14 value	Row bit, plus command encoding
`dfi_address_p0`	Row[13:0]	R0
`dfi_bank_p0`	B0	Bank select
`dfi_bg_p0`	BG0	Bank group, if present

After DRAM timing (tRCD) the MC issues the RD command at T4, phase p0:

RD command driven at T4/p0

Signal	Value	Comment
`dfi_cs_p0`	0	Chip select asserted
`dfi_act_n_p0`	1	Non-ACT form
`dfi_ras_n_p0`	1	RAS high for read
`dfi_cas_n_p0`	0	CAS low for read
`dfi_we_n_p0`	1	WE high for read
`dfi_address_p0`	Column + A10=0	C0, no auto-precharge
`dfi_bank_p0`	B0	Same bank

Two DFI clocks after the read command, at T6, the MC asserts `dfi_rddata_en` for the expected burst duration (BL8 = 8 DRAM clocks = 2 DFI clocks at 1:4):

Read enable assertion for BL8 at 1:4 ratio

Clock	dfi_rddata_en_p0	dfi_rddata_en_p1	dfi_rddata_en_p2	dfi_rddata_en_p3
T6	1	1	1	1
T7	1	1	1	1

Three DFI clocks after `dfi_rddata_en` asserts, at T9, the PHY returns `dfi_rddata_valid` and `dfi_rddata`:

Read data return for BL8

Clock	dfi_rddata_valid	dfi_rddata_w0	dfi_rddata_w1	dfi_rddata_w2	dfi_rddata_w3
T9	1	D0	D1	D2	D3
T10	1	D4	D5	D6	D7

The read data words D0-D7 correspond to the 8-word burst starting at column C0.

14.3 Walkthrough 2: activate -> write with mask

The MC wants to write to bank B1, row R0, column C0, masking byte 0 of the first data beat.

At T0 the MC issues ACT for B1/R0 as in Walkthrough 1, with `dfi_bank_p0` = B1.

After tRCD, at T4 the MC issues WR:

WR command driven at T4/p0

Signal	Value	Comment
`dfi_cs_p0`	0	Chip select asserted
`dfi_act_n_p0`	1	Non-ACT form
`dfi_ras_n_p0`	1	RAS high for write
`dfi_cas_n_p0`	0	CAS low for write
`dfi_we_n_p0`	0	WE low for write
`dfi_address_p0`	Column + A10=0	C0, no auto-precharge
`dfi_bank_p0`	B1	Same bank

Two DFI clocks after the write command, at T6, the MC asserts `dfi_wrdata_en`. One DFI clock later, at T7, write data appears on `dfi_wrdata` and the mask on `dfi_wrdata_mask`:

Write data and mask for BL8

Clock	`dfi_wrdata_en`	`dfi_wrdata_w0`	`dfi_wrdata_mask_w0`
T7	1	beat 0 (byte 0 masked)	0x01
T8	1	beat 1	0x00

The mask value `0x01` means byte 0 of beat 0 is masked; the DRAM keeps its previous value for that byte.

14.4 Walkthrough 3: back-to-back reads with chip-select hold

In a single-rank system `dfi_rddata_cs` is driven inactive-high. The PHY still expects it to be valid so it can verify the data path. This walkthrough shows two back-to-back BL8 reads to the same open row.

At T0 the MC has already activated B0/R0. At T4 it issues RD C0. At T5 it issues RD C8 (next burst-aligned column). Because both reads target the same rank, `dfi_rddata_cs` remains constant.

Back-to-back reads at 1:4 ratio

Clock	Command	`dfi_rddata_en` start	`dfi_rddata_valid` start
T4	RD C0	T6	T9
T5	RD C8	T7	T10

The read enable windows for the two commands overlap. The PHY must return valid for each command independently; the `dfi_rddata_valid` windows are T9-T10 for the first read and T10-T11 for the second read. The MC's read aligner unions or frames the enable windows to capture each burst correctly.

Source: DFI Specification v4.0 sections 3.1, 3.2, 3.3, 4.2, 4.5, 4.6

15 Timing Parameter Families

DFI 4.0 timing parameters are named `dfi_t*` or simply `t*` in the spec. They are divided into families by what they constrain and by which side of the interface defines them.

15.1 Naming convention

- `t` prefix: a timing parameter.
- Middle letters identify the interface: `cmd`, `ctrl`, `phy_wr`, `phy_rd`, `rd`, `wr`, `lp`, `calvl`, `rdlvl`, `wrlvl`, `wdqlvl`, `ctrlupd`, `phyupd`, `dram_clk`, `init`, `par`, `phy_mstr`, etc.
- `_min`/`_max` suffixes: minimum or maximum bounds.
- `_type0`..`_type3`: per-class timing for update and PHY master.

15.2 Who defines each parameter

Timing parameter definition ownership

Source	What it controls	Example parameters
MC	Controller-side scheduling commitments	`tcmd_lat`, `tparin_lat`, `tctrlupd_*`, `trdlvl_resp`, `twrlvl_resp`, `tcalvl_resp`, `twdqlvl_resp`, `tlp_resp`, `tlp_wakeup`, `tinit_start`
PHY	PHY-side propagation and response delays	`tctrl_delay`, `tphy_wrlat`, `tphy_wrdata`, `tphy_rdlat`, `tdram_clk_disable`, `tinit_complete`, `tphy_paritylat`
System	Agreed timing visible to both sides	`twrdata_delay`, `trddata_en`, `tcalvl_data`, `tcalvl_strobe`, `tphy_crcmin_lat`, `tphy_crcmax_lat`

15.3 Parameter families

15.3.1 Control timing

`tcmd_lat` and `tctrl_delay` define how commands propagate from MC to DRAM.

15.3.2 Write data timing

`tphy_wrlat`, `tphy_wrdata` and `twrdata_delay` define when write data follows a write command and when it completes on the DRAM bus.

15.3.3 Read data timing

`trddata_en`, `tphy_rdlat`, `tphy_rdcslat`, `tphy_rdcsgap` define when the MC asserts read enable, when valid data returns, and how rank switches are handled.

15.3.4 Update timing

`tctrlupd_interval`, `tctrlupd_min/max`, `tphyupd_resp`, `tphyupd_type0..3` bound the update handshake.

15.3.5 Status/init timing

`tdram_clk_disable`, `tdram_clk_enable`, `tinit_start`, `tinit_complete`, `tinit_start_min`, `tinit_complete_min`, `tparin_lat`, `tphy_paritylat` bound initialization, frequency change, clock disable and parity/alert timing.

15.3.6 Training timing

Each training operation has its own `_en`, `_resp`, `_max`, `_rr`, `_ww`, `_rw` parameters plus the CA-specific `tcalvl_*` set and the write-DQ-specific `twdqlvl_*` set.

15.3.7 Low-power timing

`tlp_resp` and `tlp_wakeup` bound the low-power handshake.

15.3.8 Error timing

`terror_resp` bounds how quickly the PHY reports an error after the offending transaction.

15.3.9 PHY master timing

`tphymstr_resp`, `tphymstr_rfsh`, `tphymstr_type0..3` bound the PHY master handshake.

15.3.10 Disconnect timing

DFI 4.0 adds a pair of `t*_disconnect` and `t*_disconnect_error` parameters for every disconnectable handshake: update, training, and PHY master.

Source: DFI Specification v4.0 sections 3.0-3.14, 5.0

16 Frequency Ratio and Phasing

DFI supports three frequency ratios between the DFI clock and the DRAM clock: 1:1, 1:2 and 1:4. The `dfi_freq_ratio` signal encodes the ratio at initialization and is expected to remain constant.

16.1 Ratio definitions

DFI frequency ratios

dfi_freq_ratio	Ratio	Meaning
00	1:1	DFI clock and DRAM clock are the same frequency.
01	1:2	One DFI clock per two DRAM clocks.
10	1:4	One DFI clock per four DRAM clocks.
11	Reserved	Not used.

At 1:4, one DFI clock covers four DRAM clocks. Control and write data signals are replicated into four phase-suffixed buses (`_p0`..`_p3`). Read data is returned as four word-suffixed buses (`_w0`..`_w3`).

16.2 What ratio does to scheduling

In a 1:1 system, the MC issues one command per DFI clock. In a 1:2 system, it may issue two commands per DFI clock (one per phase). In a 1:4 system, it may issue up to four commands per DFI clock. The PHY must accept commands on any phase.

The andesite design point uses 1:4. This means a single ACT-RD or ACT-WR sequence can be packed tightly: the ACT on phase p0 of one DFI clock and the RD/WR on phase p0 of a later DFI clock once tRCD (measured in DRAM clocks) is satisfied.

16.3 Phase replication rules

- Control interface: `_pN` suffix gives the value for phase N of the DFI PHY clock.
- Write data interface: `_pN` suffix gives the write data word for phase N.
- Read data interface: `_wN` suffix gives the read data word N.
- `dfi_alert_n`: `_aN` suffix gives the value for clock cycle N.
- Phase 0 / word 0 / cycle 0 suffixes are optional.

16.4 Timing parameter units in ratio systems

For matched-frequency systems, a DFI PHY clock equals a DFI clock. For frequency-ratio systems, timing parameters that mention “DFI PHY clocks” are measured in the higher-speed PHY clock domain (one quarter of the DFI clock period at 1:4), while parameters measured in “DFI clocks” are measured in the slower DFI clock domain.

This distinction matters for `tinit_start`, `tinit_complete`, `tparin_lat` and several others. The MC must use the correct domain when scheduling.

16.5 Ratio change

A frequency change can include a ratio change. The protocol uses `dfi_init_start` and `dfi_init_complete`. The MC must drive `dfi_freq_ratio` and `dfi_frequency` to their new values while `dfi_init_start` is asserted, and the PHY must be ready to operate at the new ratio after the handshake completes.

Source: DFI Specification v4.0 sections 3.5.4, 4.9, 4.10

17 Full Timing Parameter Table

This table lists every DFI 4.0 timing parameter from the inventory, grouped by sub-interface.

Complete DFI 4.0 timing parameter inventory

Parameter	Sub-interface	Defined by	Description
`tcmd_lat`	Control	MC	DFI PHY clocks from `dfi_cs` assertion until associated CA signals are driven (LPDDR4 CS-to-CA).
`tctrl_delay`	Control	PHY	DFI clocks from a control signal change to the change reaching the DRAM interface.
`tphy_crcmax_lat`	Write Data	System	Max DFI PHY clocks from `dfi_wrdata_en` (CRC word cycle) to associated CRC error on `dfi_alert_n`.
`tphy_crcmin_lat`	Write Data	System	Min DFI PHY clocks for the same CRC error path.
`tphy_wrcsgap`	Write Data	PHY	Min additional DFI PHY clocks between commands when changing target rank on

Parameter	Sub-interface	Defined by	Description
			dfi_wrdata_cs.
tphy_wrcslat	Write Data	PHY	DFI PHY clocks from write command to assertion of associated dfi_wrdata_cs.
tphy_wrdata	Write Data	PHY	DFI PHY clocks from dfi_wrdata_en assertion to write data driven on dfi_wrdata.
tphy_wrlat	Write Data	PHY	DFI PHY clocks from write command on control interface to dfi_wrdata_en assertion.
twrdata_delay	Write Data	System	DFI clocks from dfi_wrdata_en assertion to write data transfer completion on DRAM bus.
tphy_rdcsgap	Read Data	PHY	Min additional DFI PHY clocks between commands when changing target rank on dfi_rddata_cs.
tphy_rdcslat	Read Data	PHY	DFI PHY clocks from read command to

Parameter	Sub-interface	Defined by	Description
			assertion of associated dfi_rddata_cs.
tphy_rdlat	Read Data	PHY	Max DFI PHY clocks from dfi_rddata_en assertion to each corresponding dfi_rddata_valid bit.
trddata_en	Read Data	System	DFI PHY clocks from read command on DFI to dfi_rddata_en assertion.
tctrlupd_interval	Update	MC	Max DFI clocks the MC may wait between dfi_ctrlupd_req assertions.
tctrlupd_max	Update	MC	Max DFI clocks dfi_ctrlupd_req can be asserted.
tctrlupd_min	Update	MC	Min DFI clocks dfi_ctrlupd_req must be asserted.
tphyupd_resp	Update	PHY	Max DFI clocks from dfi_phyupd_req assertion to dfi_phyupd_ack assertion.
tphyupd_type0	Update	PHY	Max DFI clocks dfi_phyupd_req

Parameter	Sub-interface	Defined by	Description
			may stay asserted after ack for type 0.
tphyupd_type1	Update	PHY	Max DFI clocks for type 1.
tphyupd_type2	Update	PHY	Max DFI clocks for type 2.
tphyupd_type3	Update	PHY	Max DFI clocks for type 3.
tdram_clk_disable	Status	PHY	DFI clocks from dfi_dram_clk_disable assertion until DRAM clock holds low at PHY-DRAM boundary.
tdram_clk_enable	Status	PHY	DFI clocks from dfi_dram_clk_disable de-assertion to first valid DRAM clock rising edge.
tinit_complete	Status	PHY	Max DFI clocks from dfi_init_start de-assertion to dfi_init_complete re-assertion during frequency change.
tinit_complete_min	Status	PHY	Min DFI clocks before dfi_init_complete can be driven after a previous command/trainin

Parameter	Sub-interface	Defined by	Description
			g event.
tinit_start	Status	MC	DFI clocks from dfi_init_start assertion until PHY must de-assert dfi_init_complete to accept frequency change.
tinit_start_min	Status	PHY	Min DFI clocks before dfi_init_start can be driven after a previous command/training event.
tparin_lat	Status	MC	DFI PHY clocks between DFI command assertion and associated dfi_parity_in.
tphy_paritylat	Status	PHY	Max DFI clocks between dfi_parity_in and associated dfi_alert_n return (min 0x4).
terror_resp	Error	PHY	Max DFI clocks from affected DFI bus transaction(s) to dfi_error assertion.
tcalvl_ca_sel	Training	PHY	In frequency-ratio systems, defines

Parameter	Sub-interface	Defined by	Description
			`dfi_calvl_ca_sel` width generated by combining per-phase PHY inputs.
`tcalvl_capture`	Training	PHY	DFI clocks after calibration command on `dfi_cs` before `dfi_calvl_capture` pulse.
`tcalvl_cc`	Training	PHY	Min DFI clocks from one calibration command to the next.
`tcalvl_cs_ca`	Training	PHY	PHY DFI clocks from `dfi_calvl_ca_sel` assertion to `dfi_cs`.
`tcalvl_data`	Training	System	Min DFI PHY clocks `dfi_calvl_data` must be held for DRAM hold time.
`tcalvl_en`	Training	PHY	Min DFI clocks from `dfi_calvl_en` assertion to `dfi_cke` de-assertion.
`tcalvl_max`	Training	MC	Max DFI clocks MC waits for `dfi_calvl_resp` after asserting

Parameter	Sub-interface	Defined by	Description
			dfi_calvl_en.
tcalvl_resp	Training	MC	Max DFI clocks from dfi_calvl_req to MC asserting dfi_calvl_en.
tcalvl_strobe	Training	System	Min DFI PHY clocks from dfi_calvl_data to dfi_calvl_strobe for DRAM setup time.
trdlvl_en	Training	PHY	Min DFI clocks from dfi_rdlvl_en/dfi_rdlvl_gate_en assertion to first read/MRR command.
trdlvl_max	Training	MC	Max DFI clocks MC waits for dfi_rdlvl_resp after setting read training enable.
trdlvl_resp	Training	MC	Max DFI clocks from read training request to MC asserting corresponding enable.
trdlvl_rr	Training	PHY	Min DFI clocks between consecutive read-training read

Parameter	Sub-interface	Defined by	Description
			commands.
twdqlvl_en	Training	PHY	Min DFI clocks from any dfi_wdqlvl_en assertion to first write command.
twdqlvl_max	Training	MC	Max DFI clocks MC waits for dfi_wdqlvl_resp after asserting dfi_wdqlvl_en.
twdqlvl_resp	Training	MC	Max DFI clocks from dfi_wdqlvl_req to MC asserting dfi_wdqlvl_en.
twdqlvl_rw	Training	PHY	Min DFI clocks from last read in calibration sequence to first write of next set.
twdqlvl_ww	Training	PHY	Min DFI clocks inserted between write commands during write DQ training.
twrlvl_en	Training	PHY	Min DFI clocks from dfi_wrlvl_en assertion to first dfi_wrlvl_strobe assertion.
twrlvl_max	Training	MC	Max DFI clocks MC waits for

Parameter	Sub-interface	Defined by	Description
			dfi_wrlvl_resp after setting dfi_wrlvl_en.
twrlvl_resp	Training	MC	Max DFI clocks from dfi_wrlvl_req to MC asserting dfi_wrlvl_en.
twrlvl_ww	Training	PHY	Min DFI clocks from one dfi_wrlvl_strobe assertion to the next.
tlp_resp	Low Power	MC	Max DFI clocks from low-power request assertion to dfi_lp_ack assertion (min 1, max 7).
tlp_wakeup	Low Power	MC	Target max DFI clocks that dfi_lp_ack may stay asserted after request de-asserts (min 16).
tphy_db_train_en	DB Training	PHY	PHY DFI clocks after dfi_db_train_en asserts until first training command can be issued.
tphy_db_train_resp	DB Training	PHY	Max DFI clocks from PHY DQ input to

Parameter	Sub-interface	Defined by	Description
			dfi_db_train_resp output.
tphymstr_resp	PHY Master	MC	Max DFI clocks from dfi_phymstr_req assertion to dfi_phymstr_ack assertion.
tphymstr_rfsh	PHY Master	PHY	Max DFI clocks PHY needs to forward a refresh command from MC to DRAM while master.
tphymstr_type0	PHY Master	PHY	Max DFI clocks dfi_phymstr_req may stay asserted after ack for type 0.
tphymstr_type1	PHY Master	PHY	Max for type 1.
tphymstr_type2	PHY Master	PHY	Max for type 2.
tphymstr_type3	PHY Master	PHY	Max for type 3.
tcalvl_disconnect	Disconnect	PHY	Max clocks to disconnect CA training with dfi_disconnect_error=0.
tcalvl_disconnect_error	Disconnect	PHY	Same with dfi_disconnect_error=1.
tctrlupd_disconnect	Disconnect	PHY	Max clocks to disconnect a control update.
tctrlupd_disconnect_error	Disconnect	PHY	Same with error

Parameter	Sub-interface	Defined by	Description
			disconnect.
tphyupd_disconnect	Disconnect	PHY	Max clocks to disconnect a PHY update.
tphyupd_disconnect_error	Disconnect	PHY	Same with error disconnect.
tphymstr_disconnect	Disconnect	PHY	Max clocks to disconnect a PHY master request.
tphymstr_disconnect_error	Disconnect	PHY	Same with error disconnect.
trdlvl_disconnect	Disconnect	PHY	Max clocks to disconnect read data eye training.
trdlvl_disconnect_error	Disconnect	PHY	Same with error disconnect.
trdlvl_gate_disconnect	Disconnect	PHY	Max clocks to disconnect gate training.
trdlvl_gate_disconnect_error	Disconnect	PHY	Same with error disconnect.
twdqlvl_disconnect	Disconnect	PHY	Max clocks to disconnect write DQ training.
twdqlvl_disconnect_error	Disconnect	PHY	Same with error disconnect.
twrlvl_disconnect	Disconnect	PHY	Max clocks to disconnect write leveling.
twrlvl_disconnect_error	Disconnect	PHY	Same with error disconnect.
tgeardown_delay	Geardown	PHY	Delay from dfi_geardown_en

Parameter	Sub-interface	Defined by	Description
			assertion until PHY is ready to receive commands.

17.1 Programmable parameters

Complete DFI 4.0 programmable parameter inventory

Parameter	Sub-interface	Defined by	Description
`phycrc_mode`	Write Data	PHY	`0` = CRC in MC; `1` = CRC in PHY.
`phydbi_mode`	Write/Read Data	PHY	`0` = DBI in MC; `1` = DBI in PHY.
`dfidata_bit_enable`	Status	MC/PHY	Valid data bits on `dfi_wrdata` and `dfi_rddata`.
`phyfreq_range`	Status	PHY	Number of `dfi_frequency` values supported (1-32).
`mcrdlvl_slice_group`	Training	MC	Per-slice grouping of `dfi_rdlvl_en` bits.
`phyrdlvl_slice_group`	Training	PHY	PHY-side counterpart of `mcrdlvl_slice_group`.
`phycalvl_en`	Training	PHY	`1` = DFI CA training enabled.
`phyrdlvl_en`	Training	PHY	PHY supports read data eye training.
`phyrdlvl_gate_en`	Training	PHY	PHY supports gate training.
`phywdqlvl_bst`	Training	PHY	Consecutive write bursts per write DQ training sequence.
`phywrlvl_en`	Training	PHY	PHY supports

Parameter	Sub-interface	Defined by	Description
			write leveling.
`syswrlvl_strobe_num`	Training	MC	Minimum clocks `dfi_wrlvl_strobe` asserts per request.
`syscs_state`	PHY Master	MC/PHY	Per-CS active/inactive state.
`phychannel_en`	LPDDR4 Channels	System/PHY	Enabled channels in LPDDR4 combined mode.

Source: DFI Specification v4.0 sections 3.1-3.14, Tables 6, 13, 14, 16, 17, 19, 20, 22-26

18 Low-Power Handshake and Update Mechanism

18.1 Low-power control

The low-power control interface lets the MC request that the PHY enter a low-power state. It is split into separate control and data requests (`dfi_lp_ctrl_req` and `dfi_lp_data_req`) so the two sides can power down independently.

The MC asserts a request and holds `dfi_lp_wakeup` constant while awaiting acknowledge. The PHY may acknowledge (`dfi_lp_ack` within `tlp_resp`) or ignore the request. If acknowledged, the PHY de-asserts `dfi_lp_ack` within `tlp_wakeup` cycles after the request de-asserts.

The andesite design point exposes `dfi_lp_ctrl_req` and `dfi_lp_data_req` but does not consume acknowledges; an unacknowledged request times out and reports rather than blocking. This is the same dormant-pair disposition used by its predecessor.

18.1.1 Wakeup encoding

Low-power wakeup encoding

dfi_lp_wakeup	Target max cycles
0000	16
0001	32
…	doubling each step
1110	262144
1111	Unlimited

18.2 Update mechanism

18.2.1 MC-initiated update

The MC asserts `dfi_ctrlupd_req` for between `tctrlupd_min` and `tctrlupd_max` cycles. The PHY may acknowledge with `dfi_ctrlupd_ack` before `tctrlupd_min` expires. While the request is asserted, the DFI bus stays idle.

DFI 4.0 adds a requirement that a `dfi_ctrlupd_req`/`dfi_ctrlupd_ack` handshake complete immediately before any self-refresh exit command. This guarantees a clean re-entry point.

18.2.2 PHY-initiated update

The PHY asserts `dfi_phyupd_req` and drives `dfi_phyupd_type` to select one of four time classes. The MC must acknowledge with `dfi_phyupd_ack` within `tphyupd_resp`. The PHY holds `dfi_phyupd_req` for up to `tphyupd_typeX` cycles after the acknowledge. The DFI bus idles while the request is asserted.

18.2.3 DFI idle definition

The DFI bus is considered idle when no control, write data, read data, or update activity is in progress. Some training and low-power sequences require the bus to be idle before they begin.

Source: DFI Specification v4.0 sections 3.4, 3.7, 4.7, 4.8, 4.13

19 Refresh Interaction and Frequency Change

19.1 Refresh interaction

DFI does not define refresh commands specially. A refresh is just another command on the control interface: `dfi_ras_n` and `dfi_cas_n` low, `dfi_we_n` high, with `dfi_address` and `dfi_bank` set according to the memory type. The MC is responsible for scheduling refresh at the required rate and for precharging all banks before issuing an all-bank refresh.

The PHY may request control of the bus for its own refresh-related purposes through the PHY master interface. While the PHY is master, the MC may still forward refresh commands, which the PHY relays to the DRAM within `tphymstr_rfsh` cycles.

19.1.1 Refresh during training

DFI 4.0 allows refreshes to be issued during some training sequences, provided the DRAM timing requirements are met. The MC must not issue normal read or write traffic to slices that are actively being trained.

19.2 Frequency change

DFI 4.0 reuses the init handshake (`dfi_init_start`/`dfi_init_complete`) for frequency change. The MC also uses `dfi_frequency` to tell the PHY the target frequency encoding.

19.2.1 Acknowledged frequency change

1. MC drives new `dfi_frequency` and asserts `dfi_init_start`.
2. PHY accepts by de-asserting `dfi_init_complete` within `tinit_start` cycles.
3. MC holds `dfi_init_start` until the frequency switch is complete.
4. PHY re-asserts `dfi_init_complete` within `tinit_complete` cycles after `dfi_init_start` de-asserts.

19.2.2 Not-acknowledged frequency change

If the PHY does not de-assert `dfi_init_complete` within `tinit_start` cycles, the MC must abort the request and release `dfi_init_start`. The PHY must not de-assert `dfi_init_complete` after `tinit_start` expires.

19.2.3 Frequency indicator

`dfi_frequency` is a 5-bit value supporting up to 32 encodings. The mapping from encoding to clock frequency is PHY/system-defined. The `phyfreq_range` parameter tells the MC how many encodings the PHY supports. The MC must never drive an unsupported value.

`dfi_frequency` can change when `dfi_init_start` is low and should be ignored at that time. Once `dfi_init_start` asserts, `dfi_frequency` must be legal and stable.

Source: DFI Specification v4.0 sections 3.5.5, 3.5.6, 4.10

20 Revision Deltas and the andesite Tie-In

20.1 3.1 -> 4.0 revision deltas

DFI 4.0 is the largest single revision step in the DFI family. The changes fall into several groups.

20.1.1 DDR4/LPDDR4 data-path additions

Data-path deltas

Change	DFI 4.0 handling
Read/write data chip selects	New `dfi_rddata_cs` and `dfi_wrdata_cs` signals let the PHY compensate data-path timing per rank.
Read DBI	New dedicated `dfi_rddata_dbi` bus.
Write DBI	Reuses `dfi_wrdata_mask`; no separate `dfi_wrdata_dbi` exists.
Write DQ training	New `dfi_wdqlvl_*` signals and timing parameters.

20.1.2 Command and control additions

Control deltas

Change	DFI 4.0 handling
Bank groups	`dfi_bg` added in 3.0; retained in 4.0.
ACT_n	`dfi_act_n` added in 3.0; retained.
CA parity	`dfi_parity_in` and `dfi_alert_n` retained from 3.0.
Chip ID for 3DS	`dfi_cid` retained from 3.0.
LPDDR4 SDR CA	6-bit CA mapped onto `dfi_address` over two cycles.
`tcmd_lat`	New LPDDR4 CS-to-CA delay parameter.

20.1.3 System-level additions

System-level deltas

Change	DFI 4.0 handling
PHY master interface	New `dfi_phymstr_*` handshake.
Disconnect protocol	New `dfi_disconnect_error` plus disconnect timing parameters.
Frequency indicator	New `dfi_frequency` signal and `phyfreq_range` parameter.
Geardown mode	New `dfi_geardown_en` signal and `tgeardown_delay` parameter.
LPDDR4 channels	New `phychannel_en` parameter.
DB training	New `dfi_db_train_*` interface for LRDIMM.
Inactive CS support	New `syscs_state` parameter.

20.1.4 Training changes

Training deltas

Change	DFI 4.0 handling
Per-slice read leveling	`dfi_rdlvl_en`/`req` and `dfi_rdlvl_gate_en`/`req` became per-slice; grouping via `mcrdlvl_slice_group`/`phyrdlvl_slice_group`.
Training chip-select rename	`_cs_n` suffix removed: `dfi_phy_wrlvl_cs`, `dfi_phy_rdlvl_cs`, `dfi_phy_rdlvl_gate_cs`, `dfi_phy_calvl_cs`.
PHY Independent Mode	Replaces v3.1 `dfi_phylvl_req_cs_n`/`dfi_phylvl_ack_cs_n`.
CA training additions	`dfi_calvl_ca_sel`, `dfi_calvl_data`/`done`/`result`/`strobe`, and associated timing parameters.
Write leveling strobe	Now per-slice with assert count via

Change	DFI 4.0 handling
	`syswrlvl_strobe_num`.

20.1.5 Removed signal

`dfi_data_byte_disable` was removed in DFI 4.0 and replaced by the `dfidata_bit_enable` programmable parameter.

20.2 The andesite tie-in

The andesite DDR4/LPDDR4 memory controller targets DFI 4.0 because its predecessor left the entire 3.1 -> 4.0 delta as “deliberately unimplemented.” The andesite design implements that delta.

20.2.1 Inherited surface

- The async-FIFO datapath, bubble-free, at 1:4 frequency ratio.
- `dfi_init_start`/`dfi_init_complete`, `dfi_error`/`dfi_error_info`, `dfi_reset_n`.
- `dfi_lp_ctrl_req`/`dfi_lp_data_req` presented but unacknowledged-by-default.

20.2.2 Implemented 4.0 deltas

The andesite HAS Chapter 4.1 and MAS pin table list the implemented changes:

andesite implementation of DFI 4.0 deltas

DFI 4.0 feature	andesite block	Tie-in document
`dfi_act_n`, `dfi_bg`	`cmd_formatter` (modified)	`docs/andesite_has/ch04_interfaces/01_dfi_v40.md`
`dfi_parity_in`, `dfi_alert_n`	`cmd_formatter` + new alert handling	`docs/andesite_has/ch04_interfaces/01_dfi_v40.md`
`dfi_wrdata_mask` as write DBI, `dfi_rddata_dbi`	`dfi_datapath`	`docs/andesite_has/ch04_interfaces/01_dfi_v40.md`
`dfi_geardown_en`	`init_sequencer`	`docs/andesite_has/ch04_interfaces/01_dfi_v40.md`
Training (`dfi_rdlvl_*`, `dfi_calvl_*`, `dfi_wrlvl_*`)	`training` block, `rdlvl_ifc`, `ca_train_ifc`, `wrlvl_ifc`	`docs/andesite_has/ch04_interfaces/01_dfi_v40.md`, `docs/andesite_mas/ch03_interfaces/01_dfi40_pins.md`
LPDDR4 CA bus	`cmd_formatter` LPDDR4 CA submodule	`docs/andesite_has/ch04_interfaces/01_dfi_v40.md`

20.2.3 Study corrections recorded in the area docs

The andesite HAS and MAS record several study corrections made on 2026-10-04:

- The v3.1 per-CS training handshake `dfi_phylvl_req_cs_n`/`dfi_phylvl_ack_cs_n` does not exist in DFI 4.0; it is replaced by PHY Independent Mode.
- Training chip selects dropped the `_n` suffix in 4.0.
- There is no `dfi_wrdata_dbi` pin; write DBI reuses `dfi_wrdata_mask`.
- `dfi_ca_capture` was wrong in name and direction; the correct signal is `dfi_calvl_capture` (MC-to-PHY).
- `dfi_init` is not a DFI signal; the spec uses `dfi_init_start` and `dfi_init_complete`.

20.2.4 Verification posture

The andesite verification strategy targets the DFI 4.0 boundary using the in-house CocoTB framework DFI component. The BFM is DFI 4.0-partial; gaps at the andesite design point (LPDDR4 CA map, design-point JEDEC timing CSVs, gear-down validation, per-slice read leveling, CA VREF sweep feedback) are listed in `docs/andesite_has/ch06_integration/01_verification_open.md`. The verification plan exercises init sequences, gear-down entry, CA parity, ODT, FGR, LPDDR4 CA encoding, training protocols, and the dormant low-power pair.

Source: DFI Specification v4.0 revision history and sections 3.6.8, 3.12, 3.13, 3.14; andesite HAS `ch04_interfaces/01_dfi_v40.md`, andesite MAS `ch03_interfaces/01_dfi40_pins.md`, andesite HAS `ch06_integration/01_verification_open.md`